

Ali Bennani

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Profile

Master's university student at the University of Sheffield. Proficient in Python for machine learning, Data science for scalable machine learning, Computer vision and Natural language programming.

Education

University of Sheffield Artificial Intelligence Master's, Sheffield, 2025 until 2026

- Modules: Scalable Machine Learning, Text Processing, Machine Learning and Adaptive Intelligence, Natural Language Processing, Data Science with Python, Parallel Computing with Graphical Processing Units (GPUs)

University of Liverpool Computer Science Bachelor, Liverpool, First, 2022 until 2025

- Relevant First-year modules: Intro to Programming(Python), Data Structures and Algorithms(JAVA), Object-Oriented Programming(JAVA)
- Second-year modules: Complexity of algorithms(Python), Advanced Artificial Intelligence(Python)
- Third-year modules: Computer vision(Python), Multi-agent systems, Robot perception and manipulation(Python), autonomous mobile robotics(JAVA)

Liverpool John Moores University International Foundation Programme (Engineering, Computing and Life Science) with distinction, 2021 until 2022

Lycée Français International Jean Charcot , El Jadida, 2017 until 2020

- Graduated from the Scientific department with a speciality in Mathematics

SKILLS

Programing Languages

- Python for training machine learning models on small and large datasets using scikit-learn, and PySpark for scalable machine learning on high-performance clusters. data science,
- C for parallel computing on GPUs using CUDA and CPUs using OpenMP.

High-performance Computing and Linux

- Leveraged university Linux High-performance clusters to train complex machine learning models efficiently.
- Utilised PySpark to process and model massive datasets that exceeded standard computational limits and trained different ML models by making use of the distributed aspect of PySpark to reduce computation time.
- Executed shell scripting and command-line operations to streamline AI model deployment and scale data pipelines.

Cyber Security And Computer Network

- Completed a coursework project focused on Cyber Security, analysing potential threats and developing strategies to mitigate risks. More details
- Obtained my Certified in Cyber Security certification in June 2025

AI/Machine Learning:

- Using scikit and Pyspark, I implemented pipelines to transform data and trained and tested multiple models.
- Trained and tuned various neural networks for Computer Vision such as RCNN and YOLO,
- Tuned and benchmarked LLM's such as Qwen-3.5-9B, Gemma 4-4B, LFM2.5-8B-A1B and HRM-text-1B.
- Developed and trained a novel architecture for YOLO instance segmentation during my Google DeepMind internship.

Projects

Document Question Answering Assistant

- Created a pipeline that performed context-aware chunking and tokenised documents before applying a bert style model to create a question answering pipeline.

Steering vector analysis

- Extracted steering vectors for different models based on layer activation values to learn how a model represents reasoning versus information retrieval.

EXPERIENCE

Summer Intern, Google DeepMind Research Ready, June 2025 - August 2025

- Developed a high-performing AI vision component for robotic chemistry research, training a novel YOLO model to perform liquid instance segmentation with 97% accuracy.
- Learnt domain expertise in chemistry automation by engaging with industry experts, applying these insights to interdisciplinary AI projects.

Teacher Assistant, University of Liverpool, February 2024 - May 2024

- Mentored first-year students on core concepts of object-oriented programming over 10 weeks.
- Clarified assignment requirements and provided practical, step-by-step instructions on implementing Java functions and systems.

ShelfHack, University of Sheffield

- Engineered a stateful, multi-agent system using Python, Google ADK, and Gemini 2.5 to power an immersive, AI-directed adventure game.
- Deployed the agentic architecture via FastAPI to seamlessly manage and track real-time game states and user interactions.
- Partnered effectively within a hackathon team under tight time constraints to deliver an innovative AI application, securing a winning prize.

LANGUAGES:

- English: Expert, French: Expert, Arabic: Fluent, Spanish: Fluent